# Piecewise Self-Similar Radiation-Hydrodynamics Analytic Profiles

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May 2026

## 1 Wave Front and Mass Coordinates

All physical quantities are scaled using nanoseconds (ns) for time, g/cm$^2$ for mass coordinates, g/cm$^3$ for density, MBar for pressure, km/s for velocity, and HeV for temperature.

Exact physical constants (according to SI):

| Quantity | value | SI units |
|---|---|---|
| $c$ | 299792458 | m / s |
| $k_B$ | $1.380649 \times 10^{-23}$ | J / K |
| $q$ | $1.602176634 \times 10^{-19}$ | J |
| $h$ | $6.62607015 \times 10^{-34}$ | J s |

Other fundemental quantities derived are:

| Quantity | Theoretical Expression (SI) | CGS Units |
|---|---|---|
| HeV | $\frac{q}{k_B} \cdot 100$ | K |
| $\sigma_{SB}$ | $\frac{2\pi^5 k_B^4}{15c^2h^3}$ | erg/(cm$^2$ · s · K$^4$) |
| $a$ | $\frac{4\sigma_{SB}}{c}$ | erg/(cm$^3$ · K$^4$) |

**Case 1:** $\tau = 0$

**Material model.**

| Quantity | value | CGS units |
|---|---|---|
| $f$ | $3.4 \times 10^{13}$ | erg / g |
| $f_{\text{Kelvin}}$ | $\frac{f}{\text{HeV}^{\beta}}$ | erg / g |
| $\beta$ | 1.6 | - |
| $\mu$ | 0.14 | |
| $g$ | 7200 | cm$^2$ / g |
| $g_{\text{Kelvin}}$ | $\frac{g}{\text{HeV}^{\alpha}}$ | erg / g |
| $\alpha$ | 1.5 | - |
| $\lambda$ | 0.2 | - |
| $\gamma$ | 1.25 | - |
| $r$ | 0.25 | - |
| $\rho_0$ | 19.32 | g/cm$^3$ |

### 1.1 Self Similar Constants

$$B = \frac{16\sigma g}{3\beta(rf)^{\frac{4+\alpha}{\beta}}} =_{BVAL} (2)$$

$\mathrm{n} = 4+\alpha\frac{}{\beta=\frac{55}{16}k=1-\mu=1-0.14=0.86}$

$A = T_0(rf)^{\frac{1}{\beta}} =_{AVAL_{B=\frac{16\sigma g}{3\beta(rf)^{\frac{4+\alpha}{\beta}}}=_{BVAL_{n=\frac{4+\alpha}{\beta}=\frac{55}{16}k=1-\mu=1-0.14=0.86}}}}$

$$a_1 = \frac{\mu}{\beta} = \frac{7}{80} \tag{3}$$

$$a_2 = \frac{2-\mu}{\beta} = \frac{93}{80} \tag{4}$$

$$a_3 = -\frac{2}{\beta} - \tau = -\frac{5}{4} \tag{5}$$

$$b_1 = \frac{\beta(2+\lambda) - \mu(4+\alpha)}{\beta} = \frac{55}{32} \tag{6}$$

$$b_2 = \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu(4+\alpha)}{\beta} = -\frac{723}{160} \tag{7}$$

$$b_3 = \frac{8 - 3\beta + 2\alpha}{\beta} = \frac{31}{8} \tag{8}$$

$$a = \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu(4+\alpha)}{4 + 2\lambda - 4\mu} = -\frac{245}{128} \tag{9}$$

$$b = \frac{\mu - 2}{4 + 2\lambda - 4\mu} = -\frac{31}{64} \tag{10}$$

$$c = \frac{3\mu - 2\lambda - 2 + \tau\left(2(\beta - \alpha - 4) - \beta\lambda + \mu(4+\alpha)\right)}{4 + 2\lambda - 4\mu} = -\frac{33}{64} \tag{11}$$

$$a_v = \frac{2\beta - 4 - \alpha}{2 + \lambda - 2\mu} =_{AVFRAC} (12)$$

$$b_v = \frac{-1}{2 + \lambda - 2\mu} =_{BVFRAC} (13)$$

$$c_v = \frac{1 - \tau(4 + \alpha - 2\beta)}{2 + \lambda - 2\mu} =_{CVFRAC} (14)$$

$$a_u = \frac{\beta(2+\lambda) - \mu(4+\alpha)}{4 + 2\lambda - 4\mu} =_{AUFRAC} (15)$$

$$b_u = \frac{-\mu}{4 + 2\lambda - 4\mu} =_{BUFRAC} (16)$$

$$c_u = \frac{\tau\left(\beta(2+\lambda) - \mu(4+\alpha)\right) + \mu}{4 + 2\lambda - 4\mu} =_{CUFRAC} \tag{17}$$

$$a_p = \frac{\beta\lambda + 4 + \alpha - \mu(4+\alpha)}{2 + \lambda - 2\mu} =_{APFRAC} \tag{18}$$

$$b_p = \frac{1-\mu}{2+\lambda-2\mu} =_{BPFRAC} \tag{19}$$

$$c_p = \frac{\tau\left(\beta\lambda + 4 + \alpha - \mu(4+\alpha)\right) + \mu - 1}{2+\lambda-2\mu} =_{CPFRAC} \tag{20}$$

### 1.2 Wave Coordinate Normalization Variables

The ablated mass coordinate $m_f(t)$ and shocked mass coordinate $m_s(t)$ grow as exact power laws in time:

$$m_f(t) = \xi_f A^{-a} B^{-b} t^{-c} \approx_{MFSCALE_{\left(\frac{t}{\mathrm{ns}}\right)^{\frac{33}{64}} \quad \mathrm{g/cm^2}}} \tag{21}$$

$\mathrm{m}_s(t) = m_{s0}\left(\frac{t}{\mathrm{ns}}\right)^{\frac{149}{192}} \approx_{MSSCALE_{\left(\frac{t}{\mathrm{ns}}\right)^{\frac{149}{192}} \quad \mathrm{g/cm^2}}}$

So we can define the normalized mass coordinates $y = m/m_f(t)$ and $y_s = m/m_s(t)$ directly as:

$$y = \frac{m}{m_f(t)} \approx_{INV_{MF_{m\left(\frac{t}{\mathrm{ns}}\right)^{-\frac{33}{64}}}}} \tag{22}$$

$\mathrm{y}_s = \frac{m}{m_s(t)} \approx_{INV_{MS_{m\left(\frac{t}{\mathrm{ns}}\right)^{-\frac{149}{192}}}}}$

## 2 Explicit Subsonic Ablation Profiles ($m < m_f(t)$)

**Fluid Velocity** $u(m,t)$

1. **Dimensionless Fit Function**:

$$U(y) \approx_{U0_{H_{\frac{1-y}{1+_{BUHy}}}}}$$

2. **Level 1 (Theoretical Scaling)**:

$$u(m,t) = U(y)\, A^{a_u}\, B^{b_u}\, t^{c_u} = U(y)\, A^{\frac{275}{384}}\, B^{-\frac{7}{192}}\, t^{\frac{7}{192}}$$

3. **Level 2 (Parametrized Shape)**:

$$u(m,t) \approx_{U0_{H_{\frac{1-y}{1+_{BUHy\, A^{\frac{275}{384}}\, B^{-\frac{7}{192}}\, t^{\frac{7}{192}}}}}}}$$

4. **Level 2.5 (Substitute Constants at 1 ns)**: Using physical conversions ($u_{\text{scale}} \approx_{USCALE_{HKMS}\ \text{km/s}}$):

$$u(m,t) \approx_{U0_H \frac{1-y}{1+_{BUHy \cdot USCALE_{HKMS} \left(\frac{t}{\text{ns}}\right)^{\frac{7}{192}}\ \text{km/s}}}}$$

$$\approx_{LEAD_{UH} \frac{1-y}{1+_{BUHy \left(\frac{t}{\text{ns}}\right)^{\frac{7}{192}}\ \text{km/s}}}}$$

5. **Level 3 (Fully Substituted $m,t$ function)**: Substituting $y \approx_{INV_{MF}\ m\left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}}$:

$$u(m,t) \approx_{LEAD_{UH} \frac{1-_{INV_{MF}\ m\left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}}}{1+_{DENOM_{COEFF_{UH}}\ m\left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}} \left(\frac{t}{\text{ns}}\right)^{\frac{7}{192}}\ \text{km/s}}}}$$

## Specific Volume $v(m,t)$

1. **Dimensionless Fit Function**:

$$V(y) \approx_{V0_H\ (1-y)^{DVH}}$$

2. **Level 1 (Theoretical Scaling)**:

$$v(m,t) = V(y)\, A^{a_v}\, B^{b_v}\, t^{c_v} = V(y)\, A^{-\frac{115}{96}}\, B^{-\frac{25}{48}}\, t^{\frac{25}{48}}$$

3. **Level 2 (Parametrized Shape)**:

$$v(m,t) \approx_{V0_H\ (1-y)^{DVH}\ A^{-\frac{115}{96}}\ B^{-\frac{25}{48}}\ t^{\frac{25}{48}}}$$

4. **Level 2.5 (Substitute Constants at 1 ns)**: Using physical conversions ($v_{\text{scale}} \approx_{VSCALE_{H^3}\ \text{cm/g}}$):

$$v(m,t) \approx_{V0_H\ (1-y)^{DVH \cdot VSCALE_H \left(\frac{t}{\text{ns}}\right)^{\frac{25}{48}}\ \text{cm}^3/\text{g}}}$$

$$\approx_{LEAD_{VH}\ (1-y)^{DVH} \left(\frac{t}{\text{ns}}\right)^{\frac{25}{48}}\ \text{cm}^3/\text{g}}$$

5. **Level 3 (Fully Substituted $m,t$ function)**: Substituting $y \approx_{INV_{MF}\ m\left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}}$:

$$v(m,t) \approx_{LEAD_{VH} \left(1-_{INV_{MF}\ m\left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}}\right)^{DVH} \left(\frac{t}{\text{ns}}\right)^{\frac{25}{48}}\ \text{cm}^3/\text{g}}$$

## Pressure $p(m,t)$

1. **Dimensionless Fit Function**:

$$P(y) \approx_{APH\,{}_{y}{}^{CPH} +_{BPH\,{}_{y^{CPHD}}}}$$

2. **Level 1 (Theoretical Scaling)**:

$$p(m,t) = P(y)\, A^{a_p}\, B^{b_p}\, t^{c_p} = P(y)\, A^{\frac{505}{192}}\, B^{\frac{43}{96}}\, t^{-\frac{43}{96}}$$

3. **Level 2 (Parametrized Shape)**:

$$p(m,t) \approx \left[ {}_{APH\,{}_{y}{}^{CPH} +_{BPH\,{}_{y^{CPHD}}}} \right] A^{\frac{505}{192}}\, B^{\frac{43}{96}}\, t^{-\frac{43}{96}}$$

4. **Level 2.5 (Substitute Constants at 1 ns)**: Using physical conversions ($p_{\text{scale}} \approx_{PSCALE_{H}\ \text{MBar}}$):

$$p(m,t) \approx_{PSCALE_{H}\left[ {}_{APH\,{}_{y}{}^{CPH} +_{BPH\,{}_{y^{CPHD}}}} \right] \left(\frac{t}{\text{ns}}\right)^{-\frac{43}{96}}\ \text{MBar}}$$

$$\approx \left[ {}_{LEAD_{C}OEFF_{PAH}\,{}_{y}{}^{CPH} +_{LEAD_{C}OEFF_{PBH}\,{}_{y^{CPHD}}}} \right] \left(\frac{t}{\text{ns}}\right)^{-\frac{43}{96}} \quad \text{MBar}$$

5. **Level 3 (Fully Substituted $m,t$ function)**: Substituting $y \approx_{INV_{MF}\,{}_{m\left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}}}$:

$$p(m,t) \approx \left[ {}_{LEAD_{C}OEFF_{PAH}\,\left({}_{I}{}^{NV_{MF}}{}_{m\left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}}\right)^{CPH} +_{LEAD_{C}OEFF_{PBH}\left({}_{I}{}^{NV_{MF}}{}_{m\left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}}\right)^{CPHD}}} \right] \left(\frac{t}{\text{ns}}\right)^{-\frac{43}{96}} \quad \text{MBa}$$

## Temperature $T(m,t)$

1. **Dimensionless Fit Function**:

$$T(y) \approx \Big[(1-y)(1 +_{RVAL_{y)}}\Big]^{\frac{10}{39}}$$

2. **Level 1 (Theoretical Scaling)**:

$$T(m,t) = T(y)\, T_0$$

3. **Level 2 (Parametrized Shape)**:

$$T(m,t) \approx \Big[(1-y)(1 +_{RVAL_{y)}}\Big]^{\frac{10}{39}}\ T_0$$

4. **Level 2.5 (Substitute Constants at 1 ns)**: Using physical boundary drive ($T_0 = 1.00000$ HeV):

$$T(m,t) \approx 1.00000 \left[(1-y)(1+_{RVAL_{y)}}\right]^{\frac{10}{39}} \quad \text{HeV}$$

5. **Level 3 (Fully Substituted $m, t$ function)**: Substituting $y \approx_{INV_{MF}}{}_{m\left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}}$:

$$T(m,t) \approx 1.00000 \left[\left(1-_{INV_{MF}}{}_{m\left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}}\right)\left(1+_{DENOM_T}{}_{m\left(\frac{t}{\text{ns}}\right)^{-\frac{33}{64}}}\right)\right]^{\frac{10}{39}} \quad \text{HeV}$$

**Density** $\rho(m,t)$

*Note: The subsonic density profile diverges sharply at the ablation front ($\xi \to 1$), so instead of independently fitting $\tilde{\rho}(\xi)$, we compute it explicitly using the equation of state (EOS) along with the well-behaved analytical fits for pressure $p(m,t)$ and temperature $T(m,t)$.*

1. **Level 1 (Theoretical relation)**:

$$\rho(m,t) = \left(\frac{r\, f\, T(m,t)^{\beta}}{p(m,t)}\right)^{\frac{1}{\mu-1}}$$

2. **Level 2 (CGS Parametrized form)**:

$$\rho(m,t) = \left(\frac{r\, f_{\text{Kelvin}}\, T(m,t)^{\beta}}{p(m,t)}\right)^{\frac{1}{\mu-1}}$$

3. **Level 3 (Explicit numerical relation in g/cm$^3$)**: Using $r = 0.25$, $f_{\text{Kelvin}} = 3.4 \times 10^{13}/(\text{KELVIN\_PER\_HEV})^{1.6}$, $\beta = 1.6$, and $\mu = 0.14$:

$$\rho(m,t) = \left(\frac{r\, f_{\text{Kelvin}}\, T(m,t)^{1.6}}{p(m,t)}\right)^{-\frac{1}{0.86}} \quad \text{g/cm}^3$$

# 3 Explicit Shock compressed Profiles ($m_f(t) < m < m_s(t)$)

**Fluid Velocity** $u(m,t)$

1. **Dimensionless Fit Function**:

$$U(y_s) \approx_{U0S+(_{UDIFF_{COEFF_S}})\, y_s^{DUS}}$$

2. **Level 1 (Theoretical Scaling)**:

$$u(m,t) = U(y_s)\, v_0^{\frac{1}{2}}\, p_0^{\frac{1}{2}}\, t^{\frac{\tau}{2}}$$

3. **Level 2 (Parametrized Shape)**:

$$u(m,t) \approx \left[_{U0S+(_{UDIFF_{COEFF_S}})\, y_s^{DUS}}\right] v_0^{\frac{1}{2}}\, p_0^{\frac{1}{2}}\, t^{\frac{\tau}{2}}$$

4. **Level 2.5 (Substitute Constants at 1 ns)**: Using physical conversions ($u_{\text{scale}} \approx_{USCALE_{SKMS}}{}_{\text{km/s}}$):

$$u(m,t) \approx \left[ {}_{U0S+(}{}_{UDIFF_{C}OEFF_{S})}{}_{y_s^{DUS}} \right] \cdot_{USCALE_{SKMS}}{}_{\left(\frac{t}{\text{ns}}\right)^{-\frac{43}{192}}}{}_{\text{km/s}}$$

$$\approx \left[ {}_{LEAD_{C}OEFF_{U0S}+_{LEAD_{C}OEFF_{UDIFF_{S}}}}{}_{y_s^{DUS}} \right] \left(\frac{t}{\text{ns}}\right)^{-\frac{43}{192}} \quad \text{km/s}$$

5. **Level 3 (Fully Substituted $m, t$ function)**: Substituting $y_s \approx_{INV_{M}S}{}_{m\left(\frac{t}{\text{ns}}\right)^{-\frac{149}{192}}}$:

$$u(m,t) \approx \left[ {}_{LEAD_{C}OEFF_{U0S}+_{LEAD_{U}M^{ULT_{S}}}}{}_{m}{}^{DUS}{}_{\left(\frac{t}{\text{ns}}\right)^{POWER_{UTS}}} \right] \left(\frac{t}{\text{ns}}\right)^{-\frac{43}{192}} \quad \text{km/s}$$

**Pressure $p(m,t)$**

1. **Dimensionless Fit Function**:

$$P(y_s) \approx 1.0 +_{PDIFF_{C}OEFF_{S}}{}_{y_s^{DPS}}$$

2. **Level 1 (Theoretical Scaling)**:

$$p(m,t) = P(y_s)\, p_0\, t^{\tau}$$

3. **Level 2 (Parametrized Shape)**:

$$p(m,t) \approx \left[ 1.0 +_{PDIFF_{C}OEFF_{S}}{}_{y_s^{DPS}} \right] p_0\, t^{\tau}$$

4. **Level 2.5 (Substitute Constants at 1 ns)**: Using physical boundary drive ($p_0 =_{LEAD_{C}OEFF_{P0S}}{}_{\text{MBar}}$):

$$p(m,t) \approx_{LEAD_{C}OEFF_{P0S}}{}_{\left[ {}_{1.0+_{PDIFF_{C}OEFF_{S}}}{}_{y_s^{DPS}} \right]\left(\frac{t}{\text{ns}}\right)^{-\frac{43}{96}}}{}_{\text{MBar}}$$

$$\approx \left[ {}_{LEAD_{C}OEFF_{P0S}+_{LEAD_{C}OEFF_{PDIFF_{S}}}}{}_{y_s^{DPS}} \right] \left(\frac{t}{\text{ns}}\right)^{-\frac{43}{96}} \quad \text{MBar}$$

5. **Level 3 (Fully Substituted $m, t$ function)**: Substituting $y_s \approx_{INV_{M}S}{}_{m\left(\frac{t}{\text{ns}}\right)^{-\frac{149}{192}}}$:

$$p(m,t) \approx \left[ {}_{LEAD_{C}OEFF_{P0S}+_{LEAD_{P}M^{ULT_{S}}}}{}_{m}{}^{DPS}{}_{\left(\frac{t}{\text{ns}}\right)^{POWER_{PTS}}} \right] \left(\frac{t}{\text{ns}}\right)^{-\frac{43}{96}} \quad \text{MBar}$$

**Density** $\rho(m,t)$

1. **Dimensionless Fit Function**:

$$V(y_s)^{-1} \approx_{RHOSS} {}_{y_s^{DRHO_S}}$$

2. **Level 1 (Theoretical Scaling)**:

$$\rho(m,t) = \frac{1}{v_0} V(y_s)^{-1}$$

3. **Level 2 (Parametrized Shape)**:

$$\rho(m,t) \approx \rho_0 \, \rho_s \, y_s^{DRHO_S}$$

4. **Level 2.5 (Substitute Constants at 1 ns)**: Using physical density $\rho_0 = 19.32$ g/cm$^3$:

$$\rho(m,t) \approx_{LEAD_RHO_S} {}_{y_s^{DRHO_S}} {}_{\mathrm{g/cm^3}}$$

5. **Level 3 (Fully Substituted $m, t$ function)**: Substituting $y_s \approx_{INV_MS} {}_{m\left(\frac{t}{\mathrm{ns}}\right)^{-\frac{149}{192}}}$:

$$\rho(m,t) \approx_{LEAD_RHO_MULT_S} {}_{m^{DRHO_S}} {}_{\left(\frac{t}{\mathrm{ns}}\right)^{POWER_RHO_TS}} {}_{\mathrm{g/cm^3}}$$

**Temperature** $T(m,t)$

In the shocked region, the temperature profile $T(m,t)$ is derived algebraically from the equation of state (EOS) along with the well-behaved fits for pressure $p(m,t)$ and density $\rho(m,t)$:

1. **Level 1 (Theoretical relation)**:

$$T(m,t) = \left( \frac{p(m,t)\, v(m,t)^{1-\mu}}{r\, f} \right)^{\frac{1}{\beta}} = \left( \frac{p(m,t)\, \rho(m,t)^{\mu-1}}{r\, f} \right)^{\frac{1}{\beta}}$$

2. **Level 2 (CGS Parametrized form)**:

$$T(m,t) \approx \left( \frac{P(y_s)\, p_0\, t^{\tau} \left( \rho_0\, \rho_s\, y_s^{d_\rho} \right)^{\mu-1}}{r\, f_{\mathrm{Kelvin}}} \right)^{\frac{1}{\beta}}$$

3. **Level 2.25 (Algebraic simplification of bracket)**: We pull the power $y_s^{(\mu-1)}$ out of the bracket raised to $1/\beta = 0.625$, grouping all constants:

$$T(y_s,t) = T_{\mathrm{scale}} \cdot y_s^{-0.7125\, d_\rho} \left[ 1.0 + (P_s - 1.0)\, y_s^{d_p} \right]^{0.625} t^{-\frac{215}{768}}$$

4. **Level 2.5 (Substitute Constants at 1 ns)**: Using evaluated prefactors ($T_{\text{scale\_hev}} \approx_{TCOEFF_{HEV}}{}_{\text{HeV}}$):

$$T(m,t) \approx_{TCOEFF_{HEV}} {}^{POWER_{TY}1}_{\cdot y_s} \left[ 1.0 +_{DIFF_{PS}} y_s^{POWER_{TY}2} \right]^{0.625} \left(\frac{t}{\text{ns}}\right)^{-\frac{215}{768}} \text{HeV}$$

Or in Kelvin: $T(m,t) \approx_{TCOEFF_{KELVIN}} {}^{POWER_{TY}1}_{\cdot y_s} \left[ 1.0 +_{DIFF_{PS}} y_s^{POWER_{TY}2} \right]^{0.625} \left(\frac{t}{\text{ns}}\right)^{-\frac{215}{768}} \text{Kelvin}$

5. **Level 3 (Fully Substituted $m, t$ function in HeV)**: Substituting $y_s \approx_{INV_{MS}} {}_{m\left(\frac{t}{\text{ns}}\right)^{-\frac{149}{192}}}$:

$$T(m,t) \approx_{LEAD_{TMULTS}} \left[ m^{POWER_{TMS}} \left(\frac{t}{\text{ns}}\right)^{POWER_{TTS}} +_{LEAD_{TDIFF_{MULTS}}} m^{POWER_{TMS}2} \left(\frac{t}{\text{ns}}\right)^{POWER_{TTS}2} \right]^{0.625} \left(\frac{t}{\text{ns}}\right)^{-\frac{2}{7}}$$